

Aditya B. Sasanur

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SUMMARY

Machine Learning Engineer with 3+ years of experience across applied research and production ML, including 2+ years building recommendation and ranking systems at 100M+ user scale. Specialized in distributed ML infrastructure, low-latency serving, and end-to-end research to deployment, backed by published research (RecSys, NeurIPS, KDD)

EXPERIENCE

Capital One

New York City, NY

Machine Learning Engineer

January 2026 – Present

- Designed and shipped a sequential feature compression library end-to-end, shrinking feature payloads by over 80% and cutting p99 serving latency to sub-50ms in a real-time system sustaining 6K+ TPS.
- Built multi-node distributed training and inference pipelines supporting 50B+ data points, profiling thread- and memory-level optimizations across the GPU-CPU boundary to reduce transfer overhead.
- Prototyping a Large Recommender Model applying multi-task and continual learning to unify personalized ranking across a 100M+ user product ecosystem, partnering with research stakeholders from problem formulation through offline and online evaluation.

Associate Machine Learning Engineer

August 2024 – January 2026

- Engineered Spark preprocessing pipelines across multiple platforms to transform petabytes of raw data into sequential tokenized datasets for large-scale model training.
- Built real-time ML observability infrastructure spanning distributional feature drift detection, dataset shift diagnostics, model monitoring, and automated alerting to support stable deployments and explainability.
- Designed A/B testing and offline evaluation strategy with stakeholders, applying statistical significance testing to validate behavioral signals before shipping to customer-facing systems.

Georgia Tech Research Institute

Atlanta, GA

Machine Learning Researcher

June 2023 – August 2024

- Developed and deployed HAILS, a novel probabilistic hierarchical time-series forecasting model, at a Fortune 500 company across 10K+ products, achieving 8.5% accuracy improvement overall and 23% on sparse series.
- Curated the data collection for Time-MMD, the first multimodal time-series dataset (9 domains), and MM-TSFlib, an open-source forecasting library achieving 15–40% MSE reduction over unimodal baselines.
- Led the research on explainability for time series forecasting to provide grounded interpretations on industrial predictions.

PUBLICATIONS

SeqPack: Composable Compression for Recommender Systems | *First-author, RecSys 2026*

Large Scale Hierarchical Industrial Demand Time-Series Forecasting | *Second-author, KDD 2024*

Time-MMD: A Multi-domain, Multimodal Dataset for Time Series Analysis | *NeurIPS 2024*

EDUCATION

Georgia Institute of Technology

May 2024

Master of Science in Computer Science

GPA: 4.0/4.0

Georgia Institute of Technology

August 2023

Bachelor of Science in Computer Science

GPA: 3.9/4.0

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, Scala, SQL, Golang

Scientific Computing & ML: NumPy, SciPy, scikit-learn, pandas, PyTorch, Ray, TensorFlow, Hugging Face

High-Performance & Distributed Computing: Dask, Ray, Spark, RAPIDS, Databricks, Snowflake

Infrastructure: Kubernetes, Kubeflow, Docker, AWS (S3, DynamoDB, Lambda), Google Cloud, Weights & Biases